



OneRoot AgriRadius Pro

Turning any location in India into a complete
agricultural intelligence report - from free satellite data.

How it works · why it is trustworthy & trainable · how it drives
OneRoot decisions in procurement, input stores, input sales & insurance

Sourcing decisions are made half-blind

- **No trustworthy ground picture.** Which villages grow what, how much, and when to buy - usually guesswork or costly manual surveys.
- **Data you cannot verify.** Third-party crop/soil datasets exist, but there is no way to know if they match reality on the ground.
- **Blind spots beyond crops.** Livestock, dairy catchments, feed demand and allied sectors rarely feed into planning at all.
- **Static and central.** Reports go stale; field teams cannot refresh them for a new location on demand.

What OneRoot needs

- A single, on-demand view of any belt
- Crop type, acreage, cycle and prices
- Soil and climate risk for the area
- Numbers it can trust and improve over time
- Works on a phone, even when a laptop is off

WHAT IT IS

One input in. A full intelligence report out.

- **Draw a radius or drop a point** - a village, a taluk, a proposed collection centre, or an exact lat/long.
- **Or paste a Google Maps link.** The app resolves it to the area automatically.
- **Get a complete agri profile** in one screen - built live from open satellite and government data, at no data cost.

In the report

- Land cover & cropland confidence
- Crop type: coconut, arecanut, banana, maize, paddy
- Crop cycle & cropping intensity
- Soil health, rainfall history & 16-day forecast
- Mandi prices, village rankings, allied sectors

0

Rupees data cost

10 m

Satellite detail

Any

Location in India

Built on open, planetary-scale data

Every layer is computed live on Google Earth Engine from free public satellite and government sources.

Satellite & climate

- Sentinel-2 optical (red-edge bands)
- Sentinel-1 radar - sees through cloud
- Dynamic World AI land cover (Google)
- ESA WorldCover / WorldCereal
- SoilGrids soil · SRTM terrain
- CHIRPS rainfall · ERA5 / Open-Meteo

Government & market

- 20th Livestock Census 2019 (district)
- AGMARKNET mandi prices (data.gov.in)
- Central Silk Board · Fisheries dept
- Census village boundaries
- FRUITS / Bhoomi crop survey (ground truth)

Analytics

- Vegetation & moisture indices (NDVI, NDRE, NDMI...)
- Phenology - seasonal crop pulse
- Radar canopy structure
- Random-Forest crop classifier
- Cross-dataset agreement checks

Engineered to be trustworthy

- **Measured, not modelled.** Crop vigour, rainfall and radar structure are direct observations - flagged separately from modelled layers.
- **Radar sees through monsoon cloud** - paddy and plantation detection do not go blind in the growing season.
- **Two datasets must agree.** Dynamic World and ESA WorldCereal are cross-checked; disagreement is shown as lower confidence.
- **3-year stability test** flags areas where the signal is inconsistent and should be trusted less.
- **An honest confidence panel** states plainly what to trust as fact vs. treat as a range.

The trust rule

- Trust the direction and the ranges
- Verify the edges on the ground
- Every layer labelled: fact vs. estimate
- Sources cited on screen
- Accuracy is measured, not claimed

It gets smarter with every field visit

- **A ground-truth flywheel.** Field teams log what a plot actually grows; it is stored in a shared Google Sheet.
- **The model retrains** on those labels - a Random-Forest classifier that learns your region's real crop signatures.
- **Every visit raises accuracy** for that belt - the tool becomes more OneRoot-specific over time.
- **Predictions are scored** against ground truth, so improvement is measured, not assumed.

Proven in this build

- Srirangapatna: 4,300 ac of surveyed
- coconut used as ground truth
- Exposed detector was tuned for dense
- coconut - caught only 24%
- Retuned from real data → far higher
- recall. That is the flywheel working.

Crop-type detection, by physics not guesswork

How each crop is found

- Paddy - radar flood-then-growth signature
- Coconut / arecanut - flat + persistent
- tree canopy, dry-season greenness
- Banana - dense closed canopy, high swing
- Maize - strong single monsoon NDVI pulse

Beyond a single map

- Crop cycle: 1, 2 or 3 crops/year
- Acreage per village
- Cropping intensity & vigour trend
- Which belts are rising or falling

Getting to species-level

- 10 m cannot see planting rows, so
- look-alikes are grouped honestly
- The trained classifier separates them
- as ground truth accumulates
- Coconut model already calibrated

PART TWO

How it drives OneRoot decisions

Procurement · Input stores · Input sales · Insurance · Allied sectors

DECISION 1

Procurement & sourcing

- **Find the belts.** Map where your target crops actually concentrate - coconut, arecanut, maize, banana, paddy.
- **Rank villages** by cropland acreage and a sourcing score, so field effort goes where volume is.
- **Time the buy.** Crop-cycle read shows sowing/peak/harvest - when produce will hit the market.
- **Check the price.** Live mandi prices for the commodity alongside the supply picture.

Answers it gives

- Where is the coconut / maize belt?
- Which 10 villages should we source from?
- How many acres, how many cycles?
- Is harvest weeks away or months?
- What is today's mandi price?

Input-store & collection-centre siting

- **Put stores where demand is.** Cropland density and crop mix show the true catchment of a proposed location.
- **Add the livestock lens.** District cattle, buffalo and poultry densities reveal feed and veterinary demand nearby.
- **Compare candidate sites** by village rankings and area profiles before committing capital.
- **Right-size the branch** to the acreage and farmer base it will actually serve.

Answers it gives

- Which site has the biggest catchment?
- How much cropland within 10 km?
- Is there dairy / poultry demand too?
- How many farmers does it cover?
- Store, depot, or full centre?

Input sales - fertiliser, seed, feed

- **Match inputs to the crop mix.** Area crop composition points to the right fertiliser, seed and crop-protection demand.
- **Size the feed opportunity.** Livestock & poultry counts convert to an estimated feed-demand tonnage per area.
- **Target dairy belts.** Milk-pool estimates flag KMF / Aavin / Heritage catchments for cattle-feed sales.
- **Plan by season.** Crop cycle tells you when demand for each input will peak.

Answers it gives

- What inputs will this belt buy?
- How much cattle / poultry feed/day?
- Where are the dense dairy pockets?
- When does demand spike?
- Which SKUs to stock where?

Crop insurance & risk

- **Underwrite with real signals.** Soil health, rainfall history and 16-day forecast profile the risk of an area.
- **Score crop reliability.** NDVI vigour and the 3-year stability test show how consistent yields are.
- **Support parametric cover.** Objective satellite rainfall and vegetation indices are ideal triggers for index insurance.
- **Verify claims from space.** Detect crop failure or the actual planted area behind a claim.

Answers it gives

- How risky is this belt to insure?
- Is rainfall trending dry?
- Are yields stable or erratic?
- Did the insured crop actually fail?
- What area was really planted?

Allied sectors, in the same view

Livestock & dairy

- District cattle, buffalo, poultry counts
- Area-allocated to your radius
- Estimated milk pool (dairy catchment)
- Estimated feed demand (t/day)

Aquaculture

- Fish / prawn ponds detected from
- satellite (persistent water + shape)
- A real map layer, not a table
- Pond acreage measured on demand

Sericulture & fisheries

- Raw-silk output by state (KA #1 in India)
- Inland fish production by state
- District detail drops in as data is added
- Honey / apiculture context (national)

Deployed, shareable, mobile - and free

- **Runs in the cloud.** Hosted on Streamlit Cloud - the team uses it from a browser, even when your laptop is off.
- **Shared field data.** Ground-truth and soil cards sync to a shared Google Sheet the whole team writes to.
- **Mobile and desktop** - usable in the field on a phone, or at a desk for planning.
- **Free to build and run.** Open satellite data and free service tiers - a paid plan is only needed at large scale.

Cloud

Works laptop-off

Shared

Team ground truth

Mobile

Field-ready

Free

To build & run

Why this is defensible for OneRoot

- **Free and unlimited.** No per-report data cost - run it on every belt, every season.
- **Verifiable.** Cross-checked, stability-tested, and honest about confidence - not a black box.
- **It compounds.** Every field visit trains it further; the ground-truth dataset is a moat competitors cannot copy.
- **India-specific.** Tuned to South-Indian crops, soils, mandis and allied sectors - not a generic global tool.
- **Whole-farm economy.** Crops, livestock, dairy, feed, aquaculture and prices in one decision surface.

From a map to a decision engine

- Now: crop belts, cycles, soil, rainfall, prices, livestock, allied sectors - live for KA & TN.
- Next: expand the trained classifier with field ground truth (coconut model already calibrated).
- Then: add states and crops as boundaries and labels come in - accuracy compounds with use.

OneRoot AgriRadius Pro

Trustworthy because it is measured and cross-checked. Powerful because it learns from your fields.